Prep Control (to ensure sample quality):
HEK293T Sol Proteome
***In Vitro* HHS-465 100uM 1hr rt treatment**
SILAC: Probe/Probe

For detection of active-site peptides.

***when measuring inhibitor dry stocks for DMSO solutions, do not try to measure less than 2 mg of compound. The error from measuring small amounts of compound can drastically affect the final concentrations tested in your assay.**

A-1. START HERE if you are performing inhibitor competition

1. Thaw 1 mg proteome aliquots on ice for sample and control
   *proteomes are in PBS with no protease or phosphatase inhibitors
   *proteome concentrations are 2.3 mg/mL to achieve 1 mg using volumes specified below
2. Use 427 µL of proteome from #1 for the preparation below.
3. Add 5 µL of 100x compound (1 mM solution in DMSO), and incubate 30 min, 37 ^oC
4. Follow 3 and 4 in A-2

A-2. START HERE if you are only performing probe labeling

1. Thaw 1mg proteome aliquots on ice for sample and control
   *proteomes are in PBS with no protease or phosphatase inhibitors
   *proteome concentrations are 2.3 mg/mL to achieve 1 mg using volumes specified below
2. Use 432 µL of proteome from #1 for the preparation below.
3. Add 5 µL of 100x probe (or DMSO to control), gently mix tube.
4. Incubate 1 hr, room temperature (RT)
   *The time may be adjusted based on the probe.

B. Click chemistry (use only with alkyne probe; for biotin probes go to procedure C)

*if you are making a master click chemistry mix (i.e. all compounds premixed before adding to individual samples), do not add CuSO4 until just before you are ready to add master mix to samples.

1. Create master-mix by adding:
   a. Tag: Add 10 µL of 50x (10 mM, 2 mg/410 µL in DMSO) Biotin-N_3 or desthiobiotin-N_3, giving 200 µM final.
   b. TCEP: Add 10 µL of 50x (50 mM, 7.17 mg/500 µL in H_2O) TCEP (make fresh), giving 1 mM final, vortex.
   c. Ligand: Add 33 µL of 16.7x (1.67 mM) TBTA Ligand, giving 100 µM final, vortex *1.67 mM = 20 µL 50x + 180 µL DMSO + 800 µL t-BuOH 8.55mg into 200 µL in DMSO gives 50x stock*
   d. $CuSO_4$: Add 10 µL of 50x (50 mM, 12 mg in 960 µL H_2O) $CuSO_4$ (Fisher Scientific CAS 7758-98-7), giving 1 mM final, vortex
2. Add 63 µL of master-mix to each sample for a total of 500 µL reaction volume
3. Vortex to mix and allow incubation at RT for 1 hour

C. MeOH/$CHCl_3$ extraction to remove excess probe

1. Add 2 mL (4x volume) MeOH to 8 mL/2-dram glass vial.
2. Add 0.5 mL (1x Volume) $CHCl_3$
3. Add samples into the dram vial. For SILAC samples, add light samples first followed by heavy samples.

4. Add 1.0 mL* (3x volume) H_2O to sample microfuge tubes and wash out any residual proteome. Take this 1.0 mL* of H_2O and add to the dram vial already containing MeOH/ $CHCl_3$/sample
   *for SILAC samples (1 mL total sample volume), add 1.0 mL H_2O
   *for non-SILAC samples (0.5 mL total sample volume), add 1.5 mL H_2O
5. Vortex.
6. Centrifuge 1,400 g x 3 min, 4°C; Allow dram vial to rest for 3 min or until upper layer is clear
7. Remove top and bottom layers, leaving protein interface
8. Transfer protein interface to screw-cap microcentrifuge tube (Sarstedt D-51588) in 600 µL MeOH
9. Rinse dram vial with 150 µL $CHCl_3$, and add this 150 µL $CHCl_3$ to the screw-cap microcentrifuge tube. Vortex.
10. Rinse dram vial with 600 µL H_2O, and add this 600 µL H_2O to the screw-cap microcentrifuge tube. Vortex
11. Centrifuge 1,400 g x 3 min
12. Remove top and bottom layers, leaving protein interface
13. Add 600 µL MeOH, sonicate with probe sonicator 3 pulses (output 3)
14. Centrifuge 14,000 rpm for 5 min to pellet protein, remove MeOH

D. Reduction, Alkylation

1. Re-suspend pelletin 500 µL of 6 M urea/25 mM ammonium bicarbonate (ambic)
   3.6g Urea (MW=60.06) in 1 mL 250 mM NH_4HCO_3 stock
   (MW NH_4HCO_3=79.06; 198 mg into 10 mL gives 250 mM stock, make fresh)
   Bring volume up to 10 mL with milliQ water
2. Sonicate with probe sonicator 3 pulses (output 3)
3. Add 5 µL 100x (1M, 15.4 mg/100 µL in H_2O) DTT, giving 10 mM final (make fresh)
4. Incubate 65 ºC 15 min, let cool to RT or cool briefly on ice
5. Add 40 µL 50x (0.5 M, 28 mg/300 µL in H_2O) IAA, giving 40 mM IAA (make fresh)
6. Incubate RT 30 min, in the dark

E. MeOH/$CHCl_3$ extraction to remove excess reagents from reduction/alkylation

1. Add 2 mL (4x volume) MeOH to 8 mL/2-dram glass vial.
2. Add 0.5 mL (1x Volume) $CHCl_3$
3. Add samples into the dram vial.
4. Add 1.5 mL (3x volume) H_2O to screw-cap microcentrifuge tube (Sarstedt D-51588) and wash out any residual proteome. Take this 1.5 mL of H_2O and add to the dram vial already containing MeOH/ $CHCl_3$/sample
5. Vortex.
   Centrifuge 1,400 x *g* for 3 min, 4°C; Allow dram vial to rest for 3 min or until upper layer is clear
6. Remove top and bottom layers, leaving protein interface
7. Transfer protein interface to a new screw-cap microcentrifuge tube (Sarstedt D-51588) in 600 µL MeOH
8. Rinse dram vial with 150 µL $CHCl_3$ and add this 150 µL $CHCl_3$ to the screw-cap microcentrifuge tube. Vortex.
9. Rinse dram vial with 600 µL H_2O and add this 600 µL H_2O to the screw-cap microcentrifuge tube. Vortex
10. Centrifuge 1,400 g x 3 min
11. Remove top and bottom layers, leaving protein interface

12. Add 600 µL MeOH, sonicate with probe sonicator 3 pulses (output 3)
13. Centrifuge 14,000 rpm for 5 min to pellet protein, remove MeOH

F. Trypsin digestion

1. Re-suspend protein pellet in 500 µL 25 mM ambic.
2. Sonicate with probe sonicator 3 pulses (output 3)
3. Reconstitute 20 µg of sequence-grade trypsin (1 vial) with 40 µL of 25 mM ambic.
4. Add 7.5 µg (15 µL) Trypsin/Lys-c to sample and incubate at 37ºC for 3 hrs.

G. Labeled protein enrichment and elution

1. Remove 100 µL aliquot of 50% avidin slurry per sample (in this example, 1 bed volume = 100 µL) and place in 15 mL conical. Wash 3X by adding 10 mL of PBS, centrifuging at 1,400 x *g* for 1 min and decanting.
2. Resuspend washed beads in 1 bed volume of PBS. Transfer equivalent volume (approx. 130uL) of beads to separate 15 mL conicals. Use P1000 pipette tips to allow slurry to transfer evenly ensuring the amount of beads is equal in each tube.
3. Add the 500uL digested samples to the beads; washing the screw-cap tube with 1.0 mL PBS.
4. Add PBS for a total volume of 5.5 mL per sample.
5. Incubate RT, rotating for 1 hr
6. Spin 1,400 x *g* for 1 min to pellet; remove ~90% of supernatant by decanting **ONCE**

***Do not decant more than one time!!! Your beads will become re-suspended and you will dump out all of the beads on the second pour!

Wash beads/elution

1. Wash 3X with 25 mM ambic by adding 10 mL, decanting **ONCE**, and centrifuging 1,400 x *g* for 3 min.
2. Wash 3X with LC-MS water by adding 10 mL, decanting **ONCE**, and centrifuging 1,400 x *g* for 3 min.
3. Transfer beads in 200 µL of LC-MS grade water into low-bind microcentrifuge tubes.
4. Spin 1,400 x *g* for 3 min to pellet, remove ~90% of supernatant (as much as you possibly can) by pipetting with gel tip.
5. Elute peptides by adding 150 µL of elution buffer (50% ACN, 0.1% formic acid (FA)), incubating for 3 min., then spinning 1,400 x *g* for 3 min. Wait 1 min. to let beads settle after centrifuge has stopped. Now transfer 140 µL (of the original 150 µL) of eluate to a new low-bind tube using a gel loading pipette tip. Be EXTREMELY careful. DO NOT TRANSFER BEADS.
6. Repeat the elution step (Step #5) 2X, instead starting with 140 µL of elution buffer on the 2nd and 3rd elution. Transfer all eluent to a bio-rad biospin column (to trap any beads that accidentally transferred)**,** collect in clean microcentrifuge tube by centrifugation (1,400 x *g* for 3 min)
7. Poke holes from inside towards outside of microcentrifuge cap (minimize plastic debris in sample).
8. Snap freeze peptides in liquid N2 (speeds up the drying down process) and dry down on speedvac. Do not allow liquid N2 to enter into tube.
   ***AVOID performing the speedvac drying step overnight because of peptide adsorption issues described in step 10 below**
9. Re-suspend peptides in 50 µL of mobile phase A (LC-MS water + 0.1% FA)

Stage Tip cleanup of peptides

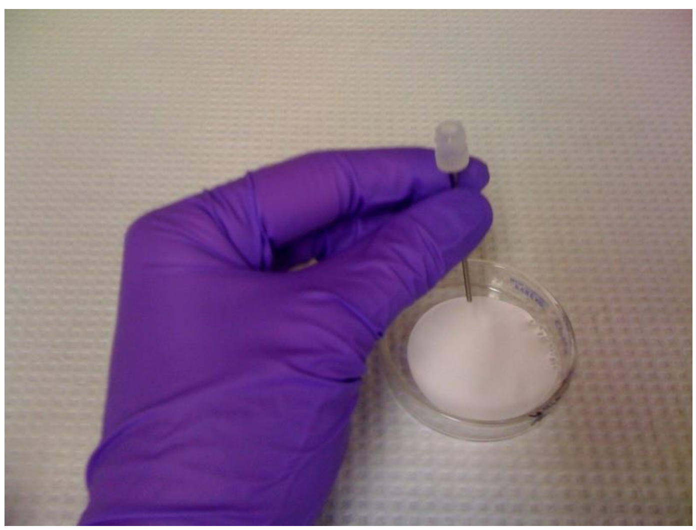
A. Protocol to prepare the StageTip

Materials:

Empore Extraction Disks:
Octadecyl (C18)-bonded silica, part no: 2215, Vendor: 3M, St. Paul, MN

<u>200 µl Pipette tips</u>
Needle: Kel-F Hub (KF), point style 3, gauge 16(1.19mm) part no: 90516
Plunger Assembly: N, RN, LT, LTN for model 1702 (25 µl)

1. Place one Empore extraction disk into a petri dish.
2. Use the Needle to pick up the stationary phase from the disk.
3. Use the plunger to insert into the 200 µl pipette tip.



B. C18-StageTips for clean-up and concentration

*** Use gel loading tips to be able to load the solvents and the sample right on top of the disk! Do not dry the disk at any time! Disk without any liquid will dry out in 1-3 min! Repeat beginning from Step 1 if you dry the disk.**

1. Condition column by wetting disk with 20 µL methanol (0.9G for 1 min)
   a. Do not let tips dry completely
2. Add 20 µL Buffer B (80% ACN + 0.1% FA) to stage tip. Spin at 900 x *g* for 1 min.
3. Add 20 µL Buffer A (Water + 0.1% FA) to stage tip. Spin at 900 x *g* for 1 min.
4. Load sample directly to the disc using gel loading tip. Spin at 900 x *g* for 1 min.
5. Wash using 20 µL Buffer A. Spin at 900 x *g* for 1 min.

*Repeat step three times. This step might take longer than 1 minute, but check it periodically and ensure the stage tip does not dry.

6. Switch to low-bind tube.
7. Elute using 20 µL buffer B. Spin at 900 x *g* for 1 min.
8. *Repeat step three times. This step might take longer than 1 minute, but check it periodically and ensure the stage tip does not dry.

9. Spike in synthetic peptide standards for normalization and QC purposes: prepare a 100 fmol/µL Angiotensin/Vasoactive (AV) standard solution and add 2.5 µL of peptide mix into each sample. *after re-suspending samples in 50 µL of mobile phase in step 12, the AV concentration in samples is 5 fmol/µL and injection of 2 µL onto the LC-MS results in analysis of 10 fmol of AV peptides.
10. Poke holes from inside towards outside of microcentrifuge cap (minimize plastic debris in sample).
11. Snap freeze peptides in liquid N2 (speeds up the drying down process) and dry down on speedvac. Do not allow liquid N2 to enter into tube.
    ***AVOID performing the speedvac drying step overnight because of peptide adsorption issues described in step 10 below**
12. Re-suspend peptides in 50 µL of mobile phase A (LC-MS water + 0.1% FA) and analyze 3 µL of sample on LC-MS.
    *note that 3 µL injection volume is generally recommended; the volume may need to be adjusted based on proteome input and protein/peptide coverage
13. **DO NOT store peptides in low-bind tubes because peptides will adsorb to the plastic and reduce signals on the LC-MS. After reconstituting peptides with mobile phase A, <u>transfer 25 µL aliquots of peptides into 2X amber vials with spring-loaded glass inserts for storage</u>. One vial is stored as backup while the other vial will allow 2X injections on the QE+ protein nanoLC-MS.**

Revised: 08/16/2022